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Triggering Presentation of an Object Based on Saliency

Abstract

In some implementations, a device includes one or more processors and a non-transitory memory. In some implementations, a method includes displaying an object within an extended reality (XR) environment, the object comprising a plurality of portions including a first portion and a second portion. In some implementations, the method includes acquiring gaze data from a user interacting with the XR environment. In some implementations, the method includes determining, based on the gaze data, a first saliency value for the first portion of the object. In some implementations, the method includes triggering presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/709,987, filed on Mar. 31, 2022, which claims the benefit of U.S. Provisional Patent App. No. 63/175,898, filed on Apr. 16, 2021, all of which are hereby incorporated by references in their entireties.

TECHNICAL FIELD

[0002] The present disclosure generally relates to triggering presentation of an object based on saliency.

BACKGROUND

[0003] Some devices are capable of generating and presenting graphical environments that include many objects. These objects may mimic real world objects. These environments may be presented on mobile communication devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] So that the present disclosure can be understood by those of ordinary skill in the art, a more detailed description may be had by reference to aspects of some illustrative implementations, some of which are shown in the accompanying drawings.

[0005] FIGS. 1A-1D are diagrams of an example operating environment in accordance with some implementations.

[0006] FIGS. 2A and 2B are a block diagrams of an object generation system in accordance with some implementations.

[0007] FIG. 3 is a flowchart representation of a method of generating an object in accordance with some implementations.

[0008] FIG. 4 is a block diagram of a device that generates an object in accordance with some implementations.

[0009] In accordance with common practice the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

SUMMARY

[0010] Various implementations disclosed herein include devices, systems, and methods for generating an object based on saliency. In some implementations, a device includes one or more processors and a non-transitory memory. In some implementations, a method includes obtaining an object that includes a plurality of portions including a first portion and a second portion. In some implementations, the method includes determining a first saliency value for the first portion of the object and a second saliency value for the second portion of the object. In some implementations, the method includes triggering presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value and the second portion of the object in accordance with a second display characteristic that is based on the second saliency value. [0011] In accordance with some implementations, a device includes one or more processors, a non-

transitory memory, and one or more programs. In some implementations, the one or more programs are stored in the non-transitory memory and are executed by the one or more processors. In some implementations, the one or more programs include instructions for performing or causing performance of any of the methods described herein. In accordance with some implementations, a non-transitory computer readable storage medium has stored therein instructions that, when executed by one or more processors of a device, cause the device to perform or cause performance of any of the methods described herein. In accordance with some implementations, a device includes one or more processors, a non-transitory memory, and means for performing or causing performance of any of the methods described herein.

DESCRIPTION

[0012] Numerous details are described in order to provide a thorough understanding of the example implementations shown in the drawings. However, the drawings merely show some example aspects of the present disclosure and are therefore not to be considered limiting. Those of ordinary skill in the art will appreciate that other effective aspects and/or variants do not include all of the specific details described herein. Moreover, well-known systems, methods, components, devices and circuits have not been described in exhaustive detail so as not to obscure more pertinent aspects of the example implementations described herein.

[0013] A physical environment refers to a physical world that people can sense and/or interact with without aid of electronic devices. The physical environment may include physical features such as a physical surface or a physical object. For example, the physical environment corresponds to a physical park that includes physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment such as through sight, touch, hearing, taste, and smell. In contrast, an extended reality (XR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an electronic device. For example, the XR environment may include augmented reality (AR) content, mixed reality (MR) content, virtual reality (VR) content, and/or the like. With an XR system, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the XR environment are adjusted in a manner that comports with at least one law of physics. As one example, the XR system may detect head movement and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. As another example, the XR system may detect movement of the electronic device presenting the XR environment (e.g., a mobile phone, a tablet, a laptop, or the like) and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations (e.g., for accessibility reasons), the XR system may adjust characteristic(s) of graphical content in the XR environment in response to representations of physical motions (e.g., vocal commands).

[0014] There are many different types of electronic systems that enable a person to sense and/or interact with various XR environments. Examples include head mountable systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones, speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head mountable system may have one or more speaker(s) and an integrated opaque display. Alternatively, a head mountable system may be configured to accept an external opaque display (e.g., a smartphone). The head mountable system may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head mountable system may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of

images is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light source, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In some implementations, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

[0015] Content creators create various objects that are rendered in graphical environments. Certain portions of an object may be salient, whereas other portions of the object may not be salient. However, when an object is rendered, a device may render the entire object with uniform display characteristics. Treating the salient portions the same as the non-salient portions tends to detract from a user experience of the device. For example, if a display resolution or a display refresh rate needs to be lowered, the entire object is displayed with a reduced resolution or refresh rate.

[0016] The present disclosure provides methods, systems, and/or devices for generating an object that is associated with saliency values that define how the object is rendered. When a content creator is creating an object, a device associates respective saliency values with different portions of the object. The saliency values trigger presentation of the different portions of the object with respective display characteristics. For example, a first portion of the object with a first saliency value is presented at a first display resolution while a second portion of the object with a second saliency value that is lower than the first saliency value is presented at a second display resolution that is less than the first display resolution. As another example, if a refresh rate of a display is being lowered in order to conserve power, a refresh rate of a first display region that is displaying the first portion may be lowered to a lesser extent than a refresh rate of a second display region that is displaying the second portion.

[0017] The device can identify the salient portions of the object automatically, for example, without receiving a user input that specifies the saliency values. The device can determine saliency values for different portions of the object based on respective amounts of time that a content creator spent creating the portions. For example, the device can set a first saliency value for a first portion of the object to be greater than a second saliency value for a second portion of the object in response to determining that the first portion was created in a first amount of time and the second portion was created in a second amount of time that is less than the first amount of time. As another example, the device can determine that portions of the object that have been revised more than a threshold number of times are more salient than portions of the object that have not been revised or that have been revised less than the threshold number of times.

[0018] The device can determine respective saliency values for different portions of an object based on corresponding characteristic values associated with the portions of the object. For example, the device can determine respective saliency values for different portions of an object based on corresponding numbers of polygons in a texture map for the object. As an example, the device can set a first saliency value for a first portion of the object based on a first number of polygons in a first portion of a texture map that corresponds to the first portion of the object, and a second saliency value for a second portion of the object based on a second number of polygons in a second portion of the texture map that corresponds to the second portion of the object. More generally, the device can determine that portions of an object that are associated with more than a threshold number of polygons in a texture map are salient, and portions of the object that are associated with less than the threshold number of polygons are non-salient.

[0019] The device can determine respective saliency values for different portions of an object based on corresponding colors of the portions of the object. As an example, the device can set a first saliency value for a first portion of the object based on a first set of colors associated with the first portion of the object, and a second saliency value for a second portion of the object based on a

second set of colors associated with the second portion of the object. More generally, the device can determine that portions of an object that utilize a first set of colors (e.g., a first number of colors, and/or a first type of colors, for example, bright colors) are salient, and portions of the object that utilize a second set of colors (e.g., a second number of colors, and/or a second type of colors, for example, grayscale or black-and-white) are non-salient.

[0020] The device can determine respective saliency values for different portions of an object based on corresponding amounts of movement that the portions of the object are capable of performing. As an example, the device can set a first saliency value for a first portion of the object based on a first amount of movement that the first portion of the object is capable of performing, and a second saliency value for a second portion of the object based on a second amount of movement that the second portion of the object is capable of performing. As another example, the device can determine that portions of an object that move are salient, and portions of the object that are stationary are non-salient.

[0021] The device can determine respective saliency values for different portions of an object based on one or more user inputs that indicate the saliency values. A user of a device (e.g., a content creator) can specify the saliency values. The content creator can identify the salient portions of the object. For example, the content creator can place pins on the salient portions of the object. The device may prompt the user to specify the salient portions of the object.

[0022] FIG. 1A is a diagram that illustrates an example physical environment **10** in accordance with some implementations. While pertinent features are shown, those of ordinary skill in the art will appreciate from the present disclosure that various other features have not been illustrated for the sake of brevity and so as not to obscure more pertinent aspects of the example implementations disclosed herein. To that end, as a non-limiting example, the physical environment **10** includes an electronic device **20** and an object generation engine **40**. In some implementations, the electronic device **20** includes a handheld computing device that can be held by a user **22**. For example, in some implementations, the electronic device **20** includes a smartphone, a tablet, a media player, a laptop, or the like. In some implementations, the electronic device **20** includes a wearable computing device that can be worn by the user **22**. For example, in some implementations, the electronic device **20** includes a head-mountable device (HMD) or an electronic watch.

[0023] In the example of FIG. 1A, the object generation engine **40** resides at the electronic device **20**. For example, the electronic device **20** implements the object generation engine **40**. In some implementations, the electronic device **20** includes a set of computer-readable instructions corresponding to the object generation engine **40**. Although the object generation engine **40** is shown as being integrated into the electronic device **20**, in some implementations, the object generation engine **40** is separate from the electronic device **20**. For example, in some implementations, the object generation engine **40** resides at another device (e.g., at a controller, a server or a cloud computing platform).

[0024] In some implementations, the object generation engine **40** presents an object generation environment **50** that allows the user **22** to generate an object such as a statue object **60**. In various implementations, an object includes various portions. In the example of FIG. 1A, the statue object **60** includes a head portion **62** (hereafter “head **62**”), a torso portion **64** (hereafter “torso **64**”) and a base portion **66** (hereafter “base **66**”). In some implementations, the object generation engine **40** generates a saliency map **70** for the statue object **60**. As illustrated in FIG. 1A, in various implementations, the saliency map **70** includes a set of one or more saliency values **72** (hereafter “saliency values **72**”) that indicate respective saliencies of corresponding portions of the statue object **60**. For example, the saliency values **72** include a head saliency value **72a** that indicates a saliency of the head **62**, a torso saliency value **72b** that indicates a saliency of the torso **64** and a base saliency value **72c** that indicates a saliency of the base **66**.

[0025] In various implementations, the object generation engine **40** associates the saliency map **70** with the statue object **60**. For example, in some implementations, the object generation engine **40**

includes the saliency map **70** in metadata associated with the statue object **60**. In some implementations, the object generation engine **40** generates procedural code that represents the statue object **60** and the saliency map **70**. In various implementations, the object generation engine **40** provides (e.g., transmits) the saliency map **70** with the statue object **60**.

[0026] In some implementations, the saliency values **72** indicate respective saliency levels of the corresponding portions of the statue object **60**. In some implementations, the saliency values **72** indicate respective levels of user interest (e.g., estimated amounts of user interest or expected amounts of interest) in the corresponding portions of the statue object **60**. For example, the head saliency value **72a** indicates a first expected amount of interest of a user in viewing the head **62** of the statue object **60**, the torso saliency value **72b** indicates a second expected amount of interest of the user in viewing the torso **64** of the statue object **60**, and the base saliency value **72c** indicates a third expected amount of interest of the user in viewing the base **66** of the statue object **60**.

[0027] In some implementations, the electronic device **20** and/or the object generation engine **40** determine the saliency values **72** based on a set of one or more user inputs **24** (hereafter “user input **24**”). In some implementations, the user input **24** includes an explicit user input. For example, in some implementations, the user input **24** specifies the saliency values **72**. In some implementations, the user input **24** includes an implicit user input. For example, in some implementations, the user input **24** includes gaze data indicating time durations for which the user **22** gazed at the portions **62**, **64** and **66** of the statue object **60**, and the saliency values **72a**, **72b** and **72c** are proportional to corresponding time durations.

[0028] In some implementations, a time duration associated with the user input **24** indicates amounts of time spent on different portions of the statue object **60**. For example, in some implementations, a time duration of the user input **24** indicates a first amount of time **26a** that the user **22** spent in creating the head **62**, a second amount of time **26b** that the user **22** spent in creating the torso **64**, and a third amount of time **26c** that the user **22** spent in creating the base **66** of the statue object **60**. In some implementations, the saliency values **72** for different portions of the statue object **60** are a function of (e.g., proportional to) corresponding amounts of time that the user **22** spent in creating the portions, for example, because the user **22** is more likely to spend more time in creating salient portions of the statue object **60** and less time in creating less salient portions of the statue object **60**. For example, in some implementations, the head saliency value **72a** is a function of the first amount of time **26a** that the user **22** spent in creating the head **62**, the torso saliency value **72b** is a function of the second amount of time **26b** that the user **22** spent in creating the torso **64**, and the base saliency value **72c** is a function of the third amount of time **26c** that the user **22** spent in creating the base **66**.

[0029] In some implementations, a number of instances of the user input **24** indicates numbers of revisions made to different portions of the statue object **60**. For example, in some implementations, a number of instances of the user input **24** indicates a first number of revisions **28a** that the user **22** made to the head **62**, a second number of revisions **28b** that the user **22** made to the torso **64**, and a third number of revisions **28c** that the user **22** made to the base **66** of the statue object **60**. In some implementations, the saliency values **72** for different portions of the statue object **60** are a function of (e.g., proportional to) corresponding numbers of revisions that the user **22** made to the portions, for example, because the user **22** is more likely to spend more time in revising more salient portions of the statue object **60** and less time in revising less salient portions of the statue object **60**. For example, in some implementations, the head saliency value **72a** is a function of the first number of revisions **28a** that the user **22** made to the head **62**, the torso saliency value **72b** is a function of the second number of revisions **28b** that the user **22** made to the torso **64**, and the base saliency value **72c** is a function of the third number of revisions **28c** that the user **22** made to the base **66**.

[0030] In some implementations, a time duration associated with the user input **24** and/or a number of instances of the user input **24** indicate amounts of detail (e.g., a number of features, for example,

a number of colors) that the user **22** incorporated into different portions of the statue object **60**. For example, in some implementations, a time duration associated with the user input **24** and/or a number of instances of the user input **24** indicate a first amount of detail **30a** that the user **22** incorporated into the head **62**, a second amount of detail **30b** that the user **22** incorporated into the torso **64**, and a third amount of detail **30c** that the user **22** incorporated into the base **66** of the statue object **60**. In some implementations, the saliency values **72** for different portions of the statue object **60** are a function of (e.g., proportional to) corresponding amounts of detail that the user **22** incorporated into the portions, for example, because the user **22** is more likely to incorporate more detail into more salient portions of the statue object **60** and less detail into less salient portions of the statue object **60**. For example, in some implementations, the head saliency value **72a** is a function of the first amount of detail **30a** that the user **22** incorporated into the head **62**, the torso saliency value **72b** is a function of the second amount of detail **30b** that the user **22** incorporated into the torso **64**, and the base saliency value **72c** is a function of the third amount of detail **30c** that the user **22** incorporated into the base **66**.

[0031] Referring to FIG. **1B**, in some implementations, the electronic device **20** and/or the object generation engine **40** determine the saliency values **72** based on characteristic values that indicate characteristics of different portions of the statue object **60**. In some implementations, the characteristic values indicate visual characteristics such as dimensions, color, etc. In some implementations, the characteristic values indicate material characteristics such as material type, texture, etc. In some implementations, the characteristic values indicate qualitative characteristics such as pixel density, polygon density in a texture map, etc. In some implementations, the characteristic values indicate quantitative characteristics such as when a portion of the statue object **60** was created.

[0032] In the example of FIG. **1B**, the head **62** is associated with a first characteristic value **62a** that indicates a characteristic of the head **62**, the torso **64** is associated with a second characteristic value **62b** that indicates a characteristic of the torso **64**, and the base **66** is associated with a third characteristic value **66a** that indicates a characteristic of the base **66**. In some implementations, the head saliency value **72a** is a function of the first characteristic value **62a**, the torso saliency value **72b** is a function of the second characteristic value **64a**, and the base saliency value **72c** is a function of the third characteristic value **66a**.

[0033] In some implementations, the saliency values **72** are a function of respective amounts of computing resources associated with the portions of the statue object **60**. In some implementations, the amounts of computing resources indicate amounts of computing resources that were utilized to generate the portions. For example, as shown in FIG. **1B**, in some implementations, a first amount of computing resources **32a** was utilized to generate the head **62**, a second amount of computing resources **32b** was utilized to generate the torso **64**, and a third amount of computing resources **32c** was utilized to generate the base **66**. In some implementations, the saliency values **72a**, **72b** and **72c** are a function of (e.g., proportional to) the amounts of computing resources **32a**, **32b** and **32c**, respectively. As such, in the example of FIG. **1B**, the head saliency value **72a** may be greater than the torso saliency value **72b** because the first amount of computing resources **32a** is greater than the second amount of computing resources **32b**, and the torso saliency value **72b** may be greater than the base saliency value **72c** because the second amount of computing resources **32b** is greater than the third amount of computing resources **32c**.

[0034] FIG. **1C** illustrates a physical environment **110** that includes an electronic device **120** that is being used by a user **122** (e.g., a content viewer), and an object rendering engine **140**. As shown in FIG. **1C**, in some implementations, the object rendering engine **140** is integrated into the electronic device **120**. Alternatively, in some implementations, the object rendering engine **140** is separate from the electronic device **120**.

[0035] In various implementations, the object rendering engine **140** presents (e.g., displays) a graphical environment such as an XR environment **150**. In some implementations, the electronic

device **120** includes a see-through display, and the XR environment **150** includes a pass-through representation of the physical environment **110**. In various implementations, the object rendering engine **140** obtains the statue object **60** and the saliency map **70**, and the object rendering engine **140** utilizes the statue object **60** and the saliency map **70** to generate and present an XR statue **160** with an XR head **162**, an XR torso **164** and an XR base **166**. In some implementations, the XR statue **160** includes a graphical object that represents the statue object **60**. As such, in some implementations, the XR statue **160** is referred to as a graphical statue.

[0036] In various implementations, the object rendering engine **140** presents (e.g., displays) different portions of the XR statue **160** with corresponding display characteristics that are based on the saliency values **72** associated with the portions. For example, as shown in FIG. **1C**, the object rendering engine **140** presents the XR head **162** with a head display characteristic **172a** that is a function of the head saliency value **72a**, the XR torso **164** with a torso display characteristic **172b** that is a function of the torso saliency value **72b**, and the XR base **166** with a base display characteristic **172c** that is a function of the base saliency value **72c**.

[0037] In some implementations, the display characteristics **172a**, **172b** and **172c** correspond to respective display resolutions. For example, the head display characteristic **172a** corresponds to a first display resolution at which the XR head **162** is displayed, the torso display characteristic **172b** corresponds to a second display resolution at which the XR torso **164** is displayed, and the base display characteristic **172c** corresponds to a third display resolution at which the XR base **166** is displayed.

[0038] In some implementations, the display characteristics **172a**, **172b** and **172c** correspond to respective refresh rates. For example, the head display characteristic **172a** corresponds to a first refresh rate at which a first set of pixels displaying the XR head **162** are refreshed, the torso display characteristic **172b** corresponds to a second refresh rate at which a second set of pixels displaying the XR torso **164** are refreshed, and the base display characteristic **172c** corresponds to a third refresh rate at which a third set of pixels displaying the XR base **166** are refreshed.

[0039] In various implementations, the display characteristics **172a**, **172b** and **172c** include numerical values (e.g., display resolutions and/or refresh rates) that are a function of the saliency values **72a**, **72b** and **72c**, respectively. For example, in some implementations, the display characteristics **172a**, **172b** and **172c** are proportional to the saliency values **72a**, **72b** and **72c**, respectively. As illustrated in FIG. **1C**, in some implementations, the head display characteristic **172a** is greater than the torso display characteristic **172b** because the head saliency value **72a** is greater than the torso saliency value **72b**, and the torso display characteristic **172b** is greater than the base display characteristic **172c** because the torso saliency value **72b** is greater than the base saliency value **72c**.

[0040] Referring to FIG. **1D**, in some implementations, the object rendering engine **140** selects a point-of-view (POV) for presenting the XR statue **160** based on the saliency values **72**. In the example of FIG. **1D**, the object rendering engine **140** is presenting the XR statue **160** from a POV in which an entirety of the XR head **162** is visible because the head saliency value **72a** is the greatest among the saliency values **72**, none of the XR base **166** is visible because the base saliency value **72c** is the smallest among the saliency values **72**, and a portion of the XR torso **164** is visible because the torso saliency value **72b** is between the head saliency value **72a** and the base saliency value **72c**. In some implementations, the object rendering engine **140** selects a POV that primarily provides a view of a portion of the XR statue **160** with the highest saliency value and does not provide a view of another portion of the XR statue **160** with the lowest saliency value. In some implementations, the object rendering engine **140** selects a POV that provides a complete view of portions with saliency values that are greater than a first threshold (e.g., the XR head **162**), a partial view of portions with saliency values that are less than the first threshold and greater than a second threshold (e.g., the XR torso **164**), and no views of portions with saliency values that are less than the second threshold (e.g., the XR base **166**).

[0041] In some implementations, the electronic device **120** includes or is attached to a head-mountable device (HMD) worn by the user **122**. The HMD presents (e.g., displays) an XR environment (e.g., the XR environment **150** shown in FIGS. **1C** and **1D**) according to various implementations. In some implementations, the HMD includes an integrated display (e.g., a built-in display) that displays the XR environment. In some implementations, the HMD includes a head-mountable enclosure. In various implementations, the head-mountable enclosure includes an attachment region to which another device with a display can be attached. For example, in some implementations, the electronic device **120** can be attached to the head-mountable enclosure. In various implementations, the head-mountable enclosure is shaped to form a receptacle for receiving another device that includes a display (e.g., the electronic device **120**). For example, in some implementations, the electronic device **120** slides/snaps into or otherwise attaches to the head-mountable enclosure. In some implementations, the display of the device attached to the head-mountable enclosure presents (e.g., displays) the XR environment. In various implementations, examples of the electronic device **120** include smartphones, tablets, media players, laptops, etc.

[0042] FIG. **2A** illustrates a block diagram of an object generation system **200** (hereafter “system **200**”) in accordance with some implementations. In some implementations, the system **200** resides at the electronic device **20** and/or the object generation engine **40** shown in FIGS. **1A** and **1B**. In some implementations, the system **200** includes an object analyzer **210**, a saliency determiner **220**, a datastore **250** and an object synthesizer **260**.

[0043] In various implementations, the object analyzer **210** obtains an object **212** (e.g., the statue object **60** shown in FIGS. **1A** and **1B**) that includes various portions (e.g., a first portion **214a**, a second portion **214b**, . . . , and an nth portion **214n**). In some implementations, the system **200** presents a user interface (e.g., the object generation environment **50**) that allows a user (e.g., a content creator, for example, the user **22** shown in FIGS. **1A** and **1B**) to create the object **212**. In some implementations, the object analyzer **210** analyzes the object **212** while the object **212** is being created. In some implementations, the object analyzer **210** analyzes the object **212** after the object **212** has been created. In some implementations, the object analyzer **210** receives the object **212** from another device. In some implementations, the object analyzer **210** retrieves the object **212** from an object datastore that stores various objects.

[0044] In some implementations, the object analyzer **210** determines respective characteristic values that indicate characteristics of the portions **214a**, **214b**, . . . , and **214n** of the object **212**. For example, as shown in FIG. **2A**, in some implementations, the object analyzer **210** determines a first characteristic value **216a** that indicates a characteristic of the first portion **214a**, a second characteristic value **216b** that indicates a characteristic of the second portion **214b**, . . . , and an nth characteristic value **216n** that indicates a characteristic of the nth portion **214n**.

[0045] In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate visual characteristics (e.g., dimensions, color, etc.) of the portions **214a**, **214b**, . . . , and **214n**, respectively. In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate material characteristics (e.g., material type, texture, etc.) of the portions **214a**, **214b**, . . . , and **214n**, respectively. In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate qualitative characteristics (e.g., pixel density, polygon density in a texture map, etc.) of the portions **214a**, **214b**, . . . , and **214n**, respectively. In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate quantitative characteristics (e.g., when a portion of the object **212** was created) of the portions **214a**, **214b**, . . . , and **214n**.

[0046] In various implementations, the saliency determiner **220** determines saliency values **240** (e.g., the saliency values **72** shown in FIGS. **1A-1D**) for different portions of the object **212**. For example, as shown in FIG. **2A**, in some implementations, the saliency determiner **220** determines a first saliency value **240a** for the first portion **214a**, a second saliency value **240b** for the second portion **214b**, . . . , and an nth saliency value **240n** for the nth portion **214n** of the object **212**.

[0047] In some implementations, the saliency determiner **220** determines the saliency values **240**

based on the characteristic values **216a**, **216b**, . . . , and **216n** provided by the object analyzer **210**. In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate respective dimensions of the portions **214a**, **214b**, . . . , and **214n**, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of (e.g., proportional to) the respective dimensions of the portions **214a**, **214b**, . . . , and **214n**. In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate respective numbers of colors of the portions **214a**, **214b**, . . . , and **214n**, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of (e.g., proportional to) the respective numbers of colors of the portions **214a**, **214b**, . . . , and **214n**.

[0048] In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate respective reflectiveness (e.g., how reflective, for example, how shiny or dull) the portions **214a**, **214b**, . . . , and **214n** are, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of (e.g., proportional to) the reflectiveness (e.g., shininess) of the portions **214a**, **214b**, . . . , and **214n**. In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate respective densities of a feature (e.g., respective pixel densities and/or respective polygon densities in a texture map of the object **212**) associated with the portions **214a**, **214b**, . . . , and **214n**, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of (e.g., proportional to) the respective densities associated with the portions **214a**, **214b**, . . . , and **214n**.

[0049] In some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate respective ages (e.g., creation dates and/or creation times) of the portions **214a**, **214b**, . . . , and **214n**, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of the respective ages of the portions **214a**, **214b**, . . . , and **214n**. For example, in some implementations, the saliency values **240** for older portions of the object **212** are greater than the saliency values **240** for newer portions of the object **212**, for example, because the older portions of the object **212** may represent a set of core building blocks of the object **212**. Alternatively, in some implementations, the saliency values **240** for newer portions of the object **212** are greater than the saliency values **240** for older portions of the object **212**, for example, because the newer portions of the object **212** may be more relevant due to their regency.

[0050] In some implementations, the saliency determiner **220** determines the saliency values **240** based on a user input **222** (e.g., the user input **24** shown in FIGS. **1A** and **1B**). In some implementations, the user input **222** includes a set of one or more user-specified saliency values **224** (hereafter “user-specified saliency values **224**”), and the saliency determiner **220** determines the saliency values **240** based on the user-specified saliency values **224**. For example, in some implementations, the saliency values **240** are within a threshold of the user-specified saliency values **224**. In some implementations, the saliency values **240** are equal to the user-specified saliency values **224**.

[0051] In some implementations, a time duration associated with the user input **222** indicates amounts of time spent on different portions of the object **212**. For example, in some implementations, a time duration of the user input **222** indicates a first amount of time **226a** that a user spent in creating the first portion **214a**, a second amount of time **226b** that the user **22** spent in creating the second portion **214b**, . . . , and an nth amount of time **226n** that the user spent in creating the nth portion **214n** of the object **212**. In some implementations, the saliency values **240** for different portions of the object **212** are a function of (e.g., proportional to) corresponding amounts of time that the user spent in creating the portions, for example, because the user is more likely to spend more time in creating salient portions of the object **212** and less time in creating less salient portions of the object **212**. For example, in some implementations, the first saliency value **240a** is a function of the first amount of time **226a** that the user spent in creating the first portion **214a** of the object **212**, the second saliency value **240b** is a function of the second amount of time **226b** that the user spent in creating the second portion **214b** of the object **212**, . . . , and the nth saliency value **240n** is a function of the nth amount of time **226n** that the user spent in creating the nth portion **214n** of the object **212**.

[0052] In some implementations, a number of instances of the user input **222** indicates numbers of revisions made to different portions of the object **212**. For example, in some implementations, a number of instances of the user input **222** indicates a first number of revisions **228a** that the user made to the first portion **214a** of the object **212**, a second number of revisions **228b** that the user made to the second portion **214b** of the object **212**, . . . , and an nth number of revisions **228n** that the user made to the nth portion **214n** of the object **212**. In some implementations, the saliency values **240** for different portions of the object **212** are a function of (e.g., proportional to) corresponding numbers of revisions that the user made to the portions, for example, because the user is more likely to spend more time in revising more salient portions of the object **212** and less time in revising less salient portions of the object **212**. For example, in some implementations, the first saliency value **240a** is a function of the first number of revisions **228a** that the user made to the first portion **214a** of the object **212**, the second saliency value **240b** is a function of the second number of revisions **228b** that the user made to the second portion **214b** of the object **212**, . . . , and the nth saliency value **240n** is a function of the nth number of revisions **228c** that the user made to the nth portion **214n** of the object **212**.

[0053] In some implementations, a time duration associated with the user input **222** and/or a number of instances of the user input **222** indicate amounts of detail (e.g., a number of features, for example, a number of colors) that the user incorporated into different portions of the object **212**. For example, in some implementations, a time duration associated with the user input **222** and/or a number of instances of the user input **222** indicate a first amount of detail **230a** that the user incorporated into the first portion **214a** of the object **212**, a second amount of detail **230b** that the user incorporated into the second portion **214b** of the object **212**, . . . , and an nth amount of detail **230c** that the user incorporated into the nth portion **214n** of the object **212**. In some implementations, the saliency values **240** for different portions of the object **212** are a function of (e.g., proportional to) corresponding amounts of detail that the user incorporated into the portions, for example, because the user is more likely to incorporate more detail into more salient portions of the object **212** and less detail into less salient portions of the object **212**. For example, in some implementations, the first saliency value **240a** is a function of the first amount of detail **230a** that the user incorporated into the first portion **214a** of the object **212**, the second saliency value **240b** is a function of the second amount of detail **230b** that the user incorporated into the second portion **214b** of the object **212**, . . . , and the nth saliency value **240n** is a function of the nth amount of detail **230n** that the user incorporated into the nth portion **214n** of the object **212**.

[0054] In some implementations, the user input **222** includes an implicit user input. For example, in some implementations, the user input **222** includes gaze data **232**, and the saliency determiner **220** determines the saliency values **240** based on the gaze data **232**. In some implementations, the gaze data **232** indicates a gaze position **232a** (e.g., pixel coordinates within the object generation environment **50** shown in FIGS. **1A** and **1B**), a gaze duration **232b** (e.g., an amount of time for which the gaze is focused on the gaze position **232a**), a gaze intensity **232c** (e.g., a dimension of the gaze, for example, a number of pixels that the gaze is directed to), and/or a gaze movement **232d** (e.g., a direction in which the gaze is moving). In some implementations, the saliency determiner **220** determines the saliency values **240** based on a combination of the gaze position **232a**, the gaze duration **232b**, the gaze intensity **232c** and/or the gaze movement **232d**. In some implementations, the saliency determiner **220** determines the saliency values **240** such that the saliency values **240** are a function of (e.g., proportional to) the gaze duration **232b**. For example, if the gaze position **232a** and the gaze duration **232b** indicate that the user **22** gazed at the second portion **214b** for a longer time duration than the remaining portions **214a**, **214c**, . . . , and **214n**, the saliency determiner **220** generates the saliency values **240** such that the second saliency value **240b** is greater than the remaining saliency values **240a**, **240c**, . . . , and **240n**.

[0055] In some implementations, the saliency determiner **220** generates the saliency values **240** based on known saliency values **252** stored in the datastore **250**. In some implementations, the

saliency determiner **220** compares the portions **214a**, **214b**, . . . , and **214n** of the object **212** with objects associated with the known saliency values **252**. In some implementations, the saliency determiner **220** sets the saliency values **240** of the portions **214a**, **214b**, . . . , and **214n** of the object **212** to known saliency values **252** of objects that are similar to the portions **214a**, **214b**, . . . , and **214n** of the object **212**. For example, in some implementations, the saliency determiner **220** sets the head saliency value **72a** (shown in FIGS. **1A-1D**) to a known saliency value **252** for a statue head. [0056] In various implementations, the object synthesizer **260** generates a saliency map **262** (e.g., the saliency map **70** shown in FIGS. **1A-1D**) that includes the saliency values **240**, and the object synthesizer **260** associates the saliency map **262** with the object **212**. For example, in some implementations, the object synthesizer **260** includes the saliency map **262** in metadata associated with the object **212**. In some implementations, the object synthesizer **260** instantiates a data container (e.g., an HTML wrapper) and includes the object **212** and the saliency map **262** in the data container. In some implementations, the object synthesizer **260** generates procedural code that represents the object **212** and the saliency map **262**. In various implementations, the object synthesizer **260** provides (e.g., transmits) the saliency map **262** with the object **212**.

[0057] Referring to FIG. **2B**, in some implementations, the object synthesizer **260** determines respective display characteristic values **264** for the portions **214a**, **214b**, . . . , and **214n** of the object **212** based on the saliency values **240**. For example, in some implementations, the object synthesizer **260** determines a first display characteristic value **264a** for the first portion **214a** of the object **212** based on the first saliency value **240a**, a second display characteristic value **264b** for the second portion **214b** of the object **212** based on the second saliency value **240b**, . . . , and an nth display characteristic value **264n** for the nth portion **214n** of the object **212** based on the nth saliency value **240n**.

[0058] In various implementations, the display characteristic values **264** indicate how corresponding portions **214a**, **214b**, . . . , and **214n** of the object **212** are to be displayed. In some implementations, the display characteristic values **264** indicate respective pixel densities for displaying the corresponding portions **214a**, **214b**, . . . , and **214n** of the object **212**. For example, the first display characteristic **264a** specifies a first pixel density for displaying the first portion **214a** of the object **212**, the second display characteristic **264b** specifies a second pixel density for displaying the second portion **214b** of the object **212**, . . . , and the nth display characteristic **264n** specifies an nth pixel density for displaying the nth portion **214n** of the object **212**.

[0059] In some implementations, the display characteristic values **264** indicate respective refresh rates for display regions of a display that displays the corresponding portions **214a**, **214b**, . . . and **214n** of the object **212**. For example, the first display characteristic **264a** specifies a first refresh rate for a first display region that displays the first portion **214a** of the object **212**, the second display characteristic **264b** specifies a second refresh rate for a second display region that displays the second portion **214b** of the object **212**, . . . , and the nth display characteristic **264n** specifies an nth refresh rate for an nth display region that displays the nth portion **214n** of the object **212**.

[0060] In some implementations, the system **200** (e.g., the object synthesizer **260**) transmits the object **212** and the display characteristic values **264** to another device, and the system **200** instructs the other device to display corresponding portions of the object **212** in accordance with respective ones of the display characteristic values **264**. For example, the system **200** instructs a device receiving the object **212** to display the first portion **214a** of the object **212** in accordance with the first display characteristic value **264a**, the second portion **214b** of the object **212** in accordance with the second display characteristic value **264b**, . . . , and the nth portion **214n** of the object **212** in accordance with the nth display characteristic value **264n**.

[0061] In some implementations, the object synthesizer **260** associates the display characteristic values **264** with the object **212** in addition to associating the saliency map **262** with the object **212**. Alternatively, in some implementations, the object synthesizer **260** associates the display characteristic values **264** as an alternative to associating the saliency map **262** with the object **212**.

In various implementations, determining the display characteristic values **264** and associating the display characteristic values **264** with the object **212** reduces a need for a device receiving the object **212** to determine the display characteristic values **264** based on the saliency values **240** included in the saliency map **262** thereby conserving computing resources at a receiving device that may be resource-constrained.

[0062] FIG. **3** is a flowchart representation of a method **300** for generating an object that is associated with different saliency values. In various implementations, the method **300** is performed by a device (e.g., the electronic device **20** and/or the object generation engine **40** shown in FIGS. **1A** and **1B**, and/or the system **200** shown in FIGS. **2A** and **2B**). In some implementations, the method **300** is performed by processing logic, including hardware, firmware, software, or a combination thereof. In some implementations, the method **300** is performed by a processor executing code stored in a non-transitory computer-readable medium (e.g., a memory).

[0063] As represented by block **310**, in various implementations, the method **300** includes obtaining an object that includes a plurality of portions including a first portion and a second portion. For example, as shown in FIGS. **1A** and **1B**, the object generation engine **40** detects that the user **22** is creating the statue object **60** with the head **62**, the torso **64** and the base **66**. As another example, as shown in FIG. **2A**, the object analyzer **210** obtains the object **212** with the portions **214a**, **214b**, . . . , and **214n**.

[0064] As represented by block **310a**, in some implementations, obtaining the object includes receiving the object. For example, as described in relation to FIG. **2A**, in some implementations, the object analyzer **210** receives the object **212** from another device that created the object **212**. In some implementations, the method **300** includes retrieving the object from a datastore that stores various objects. As such, in some implementations, the method **300** includes determining respective saliency values for different portions of the object after the object has been generated. For example, the method **300** includes determining a first saliency value for a first portion of the object, a second saliency value for a second portion of the object, . . . , and an nth saliency value for an nth portion of the object after the object has been generated.

[0065] As represented by block **310b**, in some implementations, obtaining the object includes generating the object. For example, as shown in FIGS. **1A** and **1B**, in some implementations, the object generation engine **40** presents the object generation environment **50** that allows the user **22** to generate the statue object **60**. As such, in some implementations, the method **300** includes determining respective saliency values for different portions of the object while the object is being generated. For example, the method **300** includes determining a first saliency value for a first portion of the object, a second saliency value for a second portion of the object, . . . , and an nth saliency value for an nth portion of the object while the object is being generated. In some implementations, the method **300** includes updating a saliency value for a previously-generated portion of the object as new portions of the object are generated.

[0066] As represented by block **320**, in some implementations, the method **300** includes determining a first saliency value for the first portion of the object and a second saliency value for the second portion of the object. For example, as shown in FIGS. **1A** and **1B**, the object generation engine **40** determines the head saliency value **72a** for the head **62** of the statue object **60**, the torso saliency value **72b** for the torso **64** of the statue object **60**, and the base saliency value **72c** for the base **66** of the statue object **60**. As another example, as shown in FIGS. **2A** and **2B**, the saliency determiner **220** determines the first saliency value **240a** for the first portion **214a** of the object **212**, the second saliency value **240b** for the second portion **214b** of the object **212**, . . . , and the nth saliency value **240n** for the nth portion **214n** of the object **212**.

[0067] As represented by block **320a**, in some implementations, determining the first saliency value includes determining the first saliency value based on a first characteristic value associated with the first portion of the object and determining the second saliency value includes determining the second saliency value based on a second characteristic value associated with the second portion

of the object. For example, as shown in FIG. 1B, in some implementations, the object generation engine **40** determines the head saliency value **72a** based on the first characteristic value **62a** associated with the head **62**, the torso saliency value **72b** based on the second characteristic value **62b** associated with the torso **64**, and the base saliency value **72c** based on the third characteristic value **62c** associated with the base **66**. As another example, as shown in FIGS. 2A and 2B, in some implementations, the saliency determiner **220** determines the saliency values **240a**, **240b**, . . . , and **240n** based on the characteristic values **216a**, **216b**, . . . , and **216n**, respectively.

[0068] In some implementations, the first saliency value is a function of a first amount of resources associated with the first portion of the object and the second saliency value is a function of a second amount of resources associated with the second portion of the object. In some implementations, the second amount of resources is different from the first amount of resources. In some implementations, the first saliency value is a function of a first amount of computational resources associated with (e.g., utilized in) generating the first portion of the object and the second saliency value is a function of a second amount of computational resources associated with (e.g., utilized in) generating the second portion of the object. In some implementations, the first saliency value is a function of a first amount of computational resources associated with (e.g., expected to be utilized in) presenting the first portion of the object and the second saliency value is a function of a second amount of computational resources associated with (e.g., expected to be utilized in) presenting the second portion of the object.

[0069] In some implementations, the first saliency value is a function of a first amount of time associated with generating the first portion of the object and the second saliency value is a function of a second amount of time associated with generating the second portion of the object. For example, as described in relation to FIG. 1A, in some implementations, the head saliency value **72a** is a function of the first amount of time **26a** that the user **22** spent in creating the head **62**, the torso saliency value **72b** is a function of the second amount of time **26b** that the user **22** spent in creating the torso **64**, and the base saliency value **72c** is a function of the third amount of time **26c** that the user **22** spent in creating the base **66**. As another example, as described in relation to FIGS. 2A and 2B, in some implementations, the saliency determiner **220** determines the saliency values **240a**, **240b**, . . . , and **240n** based on the amounts of time **226a**, **226b**, . . . , and **226n** associated with generating the portions **214a**, **214b**, . . . , and **214n**, respectively.

[0070] In some implementations, the first saliency value is a function of a first number of polygons in a first portion of a texture map that corresponds to the first portion of the object and the second saliency value is a function of a second number of polygons in a second portion of the texture map that corresponds to the second portion of the object. For example, as described in relation to FIG. 2A, in some implementations, the characteristic values **216a**, **216b**, . . . , and **216n** indicate respective polygon densities in a texture map of the object **212**, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of (e.g., proportional to) the respective polygon densities associated with the portions **214a**, **214b**, . . . , and **214n**.

[0071] In some implementations, the first saliency value is a function of a first level of interaction (e.g., a first level of user interaction and/or a first level of interaction by an agent) with the first portion of the object and the second saliency value is a function of a second level of interaction (e.g., a second level of user interaction and/or a second level of interaction by the agent) with the second portion of the object. In some implementations, the second level of interaction is different from the first level of interaction. In some implementations, the first saliency value is a function of a first number of revisions made to the first portion of the object and the second saliency value is a function of a second number of revisions made to the second portion of the object. For example, as described in relation to FIG. 2A, in some implementations, the saliency values **240a**, **240b**, . . . , and **240n** are a function of the respective numbers of revisions **228a**, **228b**, . . . , and **228n** that the user made to the corresponding portion **214a**, **214b**, . . . , and **214n** of the object **212**. In some implementations, the first saliency value is a function of a first gaze duration associated with the

first portion of the object (e.g., a first amount of time during which a user gazed at the first portion of the object) and the second saliency value is a function of a second gaze duration associated with the second portion of the object (e.g., a second amount of time during which the user gazed at the second portion of the object).

[0072] In some implementations, the first saliency value is a function of a first level of detail associated with (e.g., included in) the first portion of the object and the second saliency value is a function of a second level of detail associated with (e.g., included in) the second portion of the object. In some implementations, the first saliency value is a function of a first number of colors associated with the first portion of the object and the second saliency value is a function of a second number of colors associated with the second portion of the object. For example, as described in relation to FIG. 2A, in some implementations, the amounts of detail **230a**, **230b**, . . . , and **230n** indicate respective numbers of colors that the user incorporated in the corresponding portions **214a**, **214b**, . . . , and **214n**, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of the respective numbers of colors incorporated into the corresponding portions **214a**, **214b**, . . . , and **214n** of the object **212**. In some implementations, the first saliency value is a function of a first resolution associated with the first portion of the object and the second saliency value is a function of a second resolution associated with the second portion of the object.

[0073] In some implementations, the first saliency value is a function of a first amount of movement that the first portion of the object is capable of performing and the second saliency value is a function of a second amount of movement that the second portion of the object is capable of performing. As an example, referring to FIG. 2A, in some implementations, the amounts of detail **230a**, **230b**, . . . , and **230n** indicate respective amounts of movement that the corresponding portions **214a**, **214b**, . . . , and **214n** are capable of performing, and the saliency values **240a**, **240b**, . . . , and **240n** are a function of the respective amounts of movement that the portions **214a**, **214b**, . . . , and **214n** are capable of performing. In some implementations, a saliency value for a dynamic portion of the object is greater than a saliency value for a static portion of the object. In some implementations, the saliency values are a function of respective amounts of dynamism of the corresponding portions of the object. For example, a saliency value for a more dynamic portion of the object is greater than a saliency value for a less dynamic portion of the object.

[0074] As represented by block **320b**, in some implementations, determining the first saliency value and the second saliency value includes obtaining a user input indicative of the first saliency value or the second saliency value. In some implementations, the user input explicitly specifies the saliency values. For example, as shown in FIGS. 2A and 2B, in some implementations, the saliency determiner **220** obtains the user input **222** that indicates the user-specified saliency values **224**, and the saliency determiner **220** determines the saliency values **240** based on the user-specified saliency values **224**. In some implementations, the user input implicitly indicates the saliency values. For example, as shown in FIGS. 2A and 2B, in some implementations, the user input **222** indicates the amounts of time **226a**, **226b**, . . . , and **226n** associated with generating the portions **214a**, **214b**, . . . , and **214n**, the numbers of revisions **228a**, **228b**, . . . , and **228n** made to the portions **214a**, **214b**, . . . , and **214n**, the amounts of detail incorporated into the portions **214a**, **214b**, . . . , and **214n**, and/or the gaze data **232** indicative of which of the portions **214a**, **214b**, . . . , and **214n** the user gazed at and for how long.

[0075] As represented by block **330**, in some implementations, the method **300** includes triggering presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value and the second portion of the object in accordance with a second display characteristic that is based on the second saliency value. For example, as shown in FIG. 1C, in some implementations, the object generation engine **40** (shown in FIGS. 1A and 1B) triggers the XR head **162** representing the head **62** of the statue object **60** to be displayed in accordance with the head display characteristic **172a**, the XR torso **164** representing the torso **64** of the statue object **60** to be displayed in accordance with torso display characteristic **172b**, and the XR base **166**

representing the base **66** of the statue object **60** to be displayed in accordance with the base display characteristic **172c**. As another example, as described in relation to FIG. 2B, in some implementations, the system **200** triggers the portions **214a**, **214b**, . . . , and **214n** of the object **212** to be displayed in accordance with the display characteristic values **264a**, **264b**, . . . , and **264n**, respectively.

[0076] In various implementations, displaying portions of the object in accordance with respective display characteristics that are based on corresponding saliency values tends to conserve resources (e.g., rendering resources and/or battery resources) associated with displaying the object by allocating resources based on the saliency values (e.g., by allocating more resources to display salient portions of the object and less resources to display non-salient portions of the object). In some implementations, displaying portions of the object in accordance with respective display characteristics that are based on corresponding saliency values tends to enhance a user experience of the device by utilizing more resources to display portions of the object that the user is more likely to gaze at and fewer resources to display portions of the object that the user is less likely to gaze at (e.g., by displaying salient portions at a higher resolution than non-salient portions and/or by displaying salient portions with a higher refresh rate than non-salient portions).

[0077] As represented by block **330a**, in some implementations, triggering the presentation includes associating the object with a saliency map that indicates the first saliency value for the first portion of the object and the second saliency value for the second portion of the object. For example, as shown in FIGS. 2A and 2B, the object synthesizer **260** associates the object **212** with the saliency map **262** that includes the saliency values **240**.

[0078] In some implementations, the object includes a plurality of pixels, and the saliency map includes respective pixel characterization vectors for the plurality of pixels. In some implementations, each pixel characterization vector includes a saliency value for a corresponding pixel. For example, a first pixel characterization vector for a first one of the plurality of pixels includes a first saliency value for the first one of the plurality of pixels, a second pixel characterization vector for a second one of the plurality of pixels includes a second saliency value for the second one of the plurality of pixels, . . . , and an nth pixel characterization vector for an nth one of the plurality of pixels includes an nth saliency value for the nth one of the plurality of pixels.

[0079] As represented by block **330b**, in some implementations, triggering the presentation includes triggering display of the first portion of the object with a first pixel density that is a function of the first saliency value and the second portion of the object with a second pixel density that is a function of the second saliency value. For example, referring to FIG. 2B, in some implementations, the display characteristic values **264a**, **264b**, . . . , and **264n** indicate respective pixel densities (e.g., respective resolutions) for the corresponding portions **214a**, **214b**, . . . , and **214n** of the object **212**.

[0080] In some implementations, triggering the presentation includes triggering display of the first portion of the object at a first refresh rate that is a function of the first saliency value and the second portion of the object at a second refresh rate that is a function of the second saliency value. For example, referring to FIG. 2B, in some implementations, the display characteristic values **264a**, **264b**, . . . , and **264n** indicate respective refresh rates for display regions of a display that displays the corresponding portions **214a**, **214b**, . . . , and **214n** of the object **212**.

[0081] As represented by block **330c**, in some implementations, triggering the presentation includes triggering display of the object from a first point-of-view (POV) that primarily focuses on the first portion of the object in response to the first saliency value being greater than the second saliency value and triggering display of the object from a second POV that primarily focuses on the second portion of the object in response to the second saliency value being greater than the first saliency value. For example, as shown in FIG. 1D, the electronic device **120** displays the XR statue **160** from a POV that shows the XR head **162** (e.g., because the head saliency value **72a** is greater than a first saliency threshold and a second saliency threshold) and a portion of the XR torso **164**

(e.g., because the torso saliency value **72b** is between the first and second saliency thresholds) and not the XR base **166** (e.g., because the base saliency value **72c** is lower than the first and second saliency thresholds).

[0082] As represented by block **330d**, in some implementations, the method **300** includes triggering an agent to interact with the first portion of the object in response to the first saliency value being greater than the second saliency value. In some implementations, interacting with the first portion includes directing a gaze of the agent towards the first portion of the object. For example, referring to FIG. **1C**, in some implementations, an XR representation of an agent is animated to gaze at the XR head **162** instead of the XR base **166** because the head saliency value **72a** is greater than the base saliency value **72c**. In some implementations, interacting with the first portion includes animating the XR representation of the agent in order to provide an appearance that the agent is touching the first portion of the object. For example, referring to FIG. **1C**, in some implementations, an XR representation of an agent is animated to touch the XR head **162** instead of the XR base **166** because the head saliency value **72a** is greater than the base saliency value **72c**.

[0083] FIG. **4** is a block diagram of a device **400** in accordance with some implementations. In some implementations, the device **400** implements the electronic device **20** and/or the object generation engine **40** shown in FIGS. **1A** and **1B**, the electronic device **120** and/or the object rendering engine **140** shown in FIGS. **1C** and **1D**, and/or the system **200** shown in FIGS. **2A** and **2B**. While certain specific features are illustrated, those of ordinary skill in the art will appreciate from the present disclosure that various other features have not been illustrated for the sake of brevity, and so as not to obscure more pertinent aspects of the implementations disclosed herein. To that end, as a non-limiting example, in some implementations the device **400** includes one or more processing units (CPUs) **401**, a network interface **402**, a programming interface **403**, a memory **404**, one or more input/output (I/O) devices **410**, and one or more communication buses **405** for interconnecting these and various other components.

[0084] In some implementations, the network interface **402** is provided to, among other uses, establish and maintain a metadata tunnel between a cloud hosted network management system and at least one private network including one or more compliant devices. In some implementations, the one or more communication buses **405** include circuitry that interconnects and controls communications between system components. The memory **404** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. The memory **404** optionally includes one or more storage devices remotely located from the one or more CPUs **401**. The memory **404** comprises a non-transitory computer readable storage medium.

[0085] In some implementations, the memory **404** or the non-transitory computer readable storage medium of the memory **404** stores the following programs, modules and data structures, or a subset thereof including an optional operating system **406**, the object analyzer **210**, the saliency determiner **220** and the object synthesizer **260**. In various implementations, the device **400** performs the method **300** shown in FIG. **3**.

[0086] In some implementations, the object analyzer **210** includes instructions **210a**, and heuristics and metadata **210b** for obtaining an object that includes a plurality of portions including a first portion and a second portion. In some implementations, the object analyzer **210** performs at least some of the operation(s) represented by block **310** in FIG. **3**.

[0087] In some implementations, the saliency determiner **220** includes instructions **220a**, and heuristics and metadata **220b** for determining a first saliency value for the first portion of the object and a second saliency value for the second portion of the object. In some implementations, the saliency determiner **220** performs at least some of the operation(s) represented by block **320** in FIG. **3**.

[0088] In some implementations, the object synthesizer **260** instructions **260a**, and heuristics and

metadata **260b** for triggering presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value and the second portion of the object in accordance with a second display characteristic that is based on the second saliency value. In some implementations, the object synthesizer **260** performs at least some of the operation(s) represented by block **330** in FIG. 3.

[0089] In some implementations, the one or more I/O devices **410** include an input device for obtaining user inputs (e.g., the user input **24** shown in FIG. 1A and/or the user input **222** shown in FIGS. 2A and 2B). In some implementations, the input device includes a touchscreen, a depth sensor (e.g., a depth camera) and/or an image sensor (e.g., a camera, for example, a visible light camera or an infrared light camera). In some implementations, the one or more I/O devices **410** include a display for displaying an object (e.g., the statue object **60** shown in FIGS. 1A and 1B, the XR statue **160** shown in FIGS. 1C and 1D, and/or the object **212** shown in FIGS. 2A and 2B).

[0090] In various implementations, the one or more I/O devices **410** include a video pass-through display which displays at least a portion of a physical environment surrounding the device **400** as an image captured by a scene camera. In various implementations, the one or more I/O devices **410** include an optical see-through display which is at least partially transparent and passes light emitted by or reflected off the physical environment.

[0091] It will be appreciated that FIG. 4 is intended as a functional description of the various features which may be present in a particular implementation as opposed to a structural schematic of the implementations described herein. As recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated. For example, some functional blocks shown separately in FIG. 4 could be implemented as a single block, and the various functions of single functional blocks could be implemented by one or more functional blocks in various implementations. The actual number of blocks and the division of particular functions and how features are allocated among them will vary from one implementation to another and, in some implementations, depends in part on the particular combination of hardware, software, and/or firmware chosen for a particular implementation.

[0092] While various aspects of implementations within the scope of the appended claims are described above, it should be apparent that the various features of implementations described above may be embodied in a wide variety of forms and that any specific structure and/or function described above is merely illustrative. Based on the present disclosure one skilled in the art should appreciate that an aspect described herein may be implemented independently of any other aspects and that two or more of these aspects may be combined in various ways. For example, an apparatus may be implemented and/or a method may be practiced using any number of the aspects set forth herein. In addition, such an apparatus may be implemented and/or such a method may be practiced using other structure and/or functionality in addition to or other than one or more of the aspects set forth herein.

Claims

1. A method comprising: at a device including one or more processors and a non-transitory memory: displaying an object within an extended reality (XR) environment, the object comprising a plurality of portions including a first portion and a second portion; acquiring gaze data from a user interacting with the XR environment; determining, based on the gaze data, a first saliency value for the first portion of the object; and triggering presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value.
2. The method of claim 1, further comprising: triggering an agent within the XR environment to direct a gaze toward the first portion of the object in response to determining that the first saliency value exceeds a predefined threshold.
3. The method of claim 1, further comprising: animating a representation of an agent within the XR

environment to provide a visual appearance that the agent is touching the first portion of the object, wherein the animation is triggered based on the first saliency value.

4. The method of claim 1, further comprising: determining whether the first saliency value exceeds a threshold, and in response: when the threshold is exceeded, animating an agent to interact with the first portion; and when the threshold is not exceeded, suppressing or deferring interaction with the first portion.

5. The method of claim 1, wherein the first saliency value is a function of a first time duration for which a user gazed at the first portion of the object.

6. The method of claim 5, wherein the first saliency value is proportional to the first time duration.

7. The method of claim 1, wherein the first saliency value is a function of a first gaze position within the XR environment.

8. The method of claim 1, wherein the first saliency value is a function of a first gaze intensity indicating a dimension associated with a gaze corresponding to the gaze data.

9. The method of claim 8, wherein the dimension comprises a number of pixels that the gaze is directed to.

10. The method of claim 1, wherein the first saliency value is a function of a first gaze movement indicating a direction of gaze motion.

11. The method of claim 1, further comprising determining, based on the gaze data, a second saliency value for the second portion of the object, wherein the second saliency value is used to determine a second display characteristic for presentation of the second portion.

12. The method of claim 11, wherein the first saliency value is a function of a first amount of resources associated with the first portion of the object and the second saliency value is a function of a second amount of resources associated with the second portion of the object.

13. The method of claim 11, wherein the first saliency value is a function of a first amount of time associated with generating the first portion of the object and the second saliency value is a function of a second amount of time associated with generating the second portion of the object.

14. The method of claim 11, wherein the first saliency value is a function of a first number of polygons in a first portion of a texture map that corresponds to the first portion of the object and the second saliency value is a function of a second number of polygons in a second portion of the texture map that corresponds to the second portion of the object.

15. The method of claim 11, wherein the first saliency value is a function of a first level of interaction with the first portion of the object and the second saliency value is a function of a second level of interaction with the second portion of the object.

16. The method of claim 11, wherein the first saliency value is a function of a first number of revisions made to the first portion of the object and the second saliency value is a function of a second number of revisions made to the second portion of the object.

17. The method of claim 11, wherein the first saliency value is a function of a first level of detail associated with the first portion of the object and the second saliency value is a function of a second level of detail associated with the second portion of the object.

18. The method of claim 11, wherein triggering the presentation comprises associating the object with a saliency map that indicates the first saliency value for the first portion of the object and the second saliency value for the second portion of the object.

19. A device comprising: one or more processors; a non-transitory memory; and one or more programs stored in the non-transitory memory, which, when executed by the one or more processors, cause the device to: display an object within an extended reality (XR) environment, the object comprising a plurality of portions including a first portion and a second portion; acquire gaze data from a user interacting with the XR environment; determine, based on the gaze data, a first saliency value for the first portion of the object; and trigger presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value.

20. A non-transitory memory storing one or more programs, which, when executed by one or more

processors of a device, cause the device to: display an object within an extended reality (XR) environment, the object comprising a plurality of portions including a first portion and a second portion; acquire gaze data from a user interacting with the XR environment; determine, based on the gaze data, a first saliency value for the first portion of the object; and trigger presentation of the first portion of the object in accordance with a first display characteristic that is based on the first saliency value.
